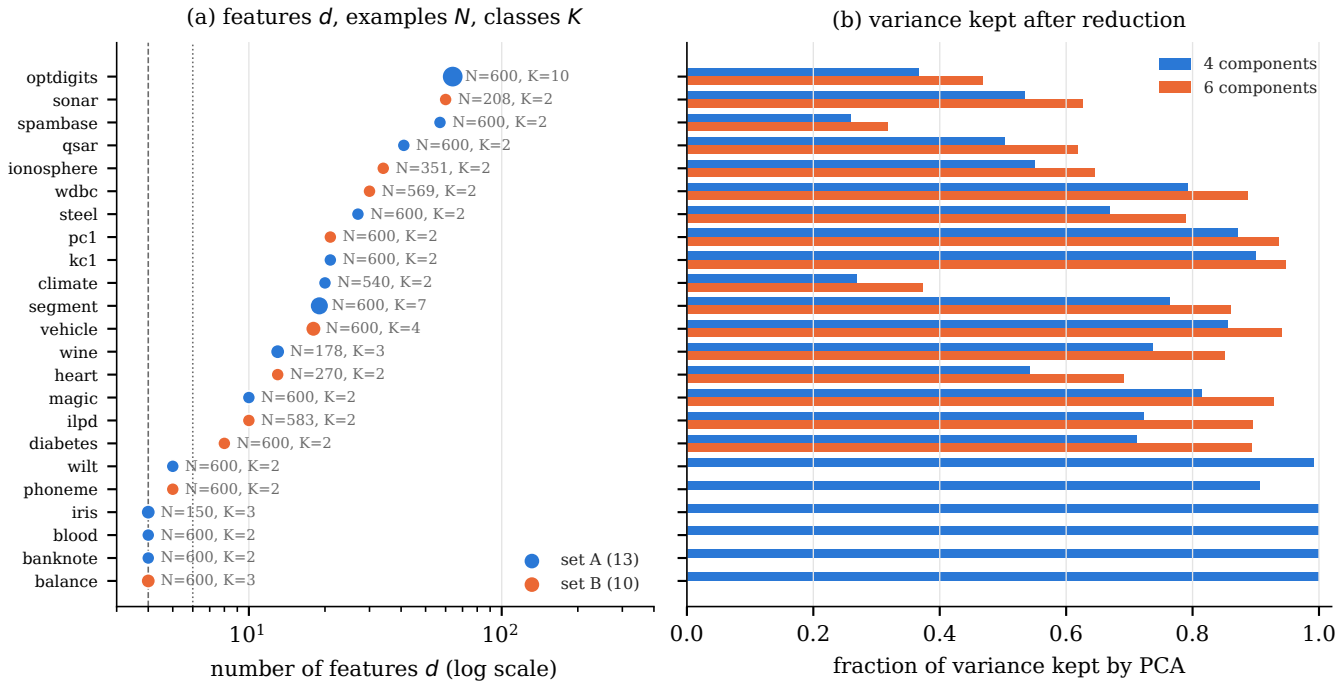
(a) features d, examples N, classes K
optdigits
sonar
spambase
qsar
ionosphere
wdbc
steel
pc1
kc1
climate
segment
vehicle
wine
heart
magic
ilpd
diabetes
wilt
phoneme
iris
blood
banknote
balance
N=600, K=10
N=208, K=2
N=600, K=2
N=600, K=2
N=351, K=2
N=569, K=2
N=600, K=2
N=600, K=2
N=600, K=2
N=540, K=2
N=600, K=7
N=600, K=4
N=178, K=3
N=270, K=2
N=600, K=2
N=583, K=2
N=600, K=2
N=600, K=2
N=600, K=2
N=150, K=3
N=600, K=2
N=600, K=2
N=600, K=3
set A (13)
set B (10)
10^1
10^2
number of features d (log scale)
(b) variance kept after reduction
4 components
6 components
0.0
0.2
0.4
0.6
0.8
1.0
fraction of variance kept by PCA